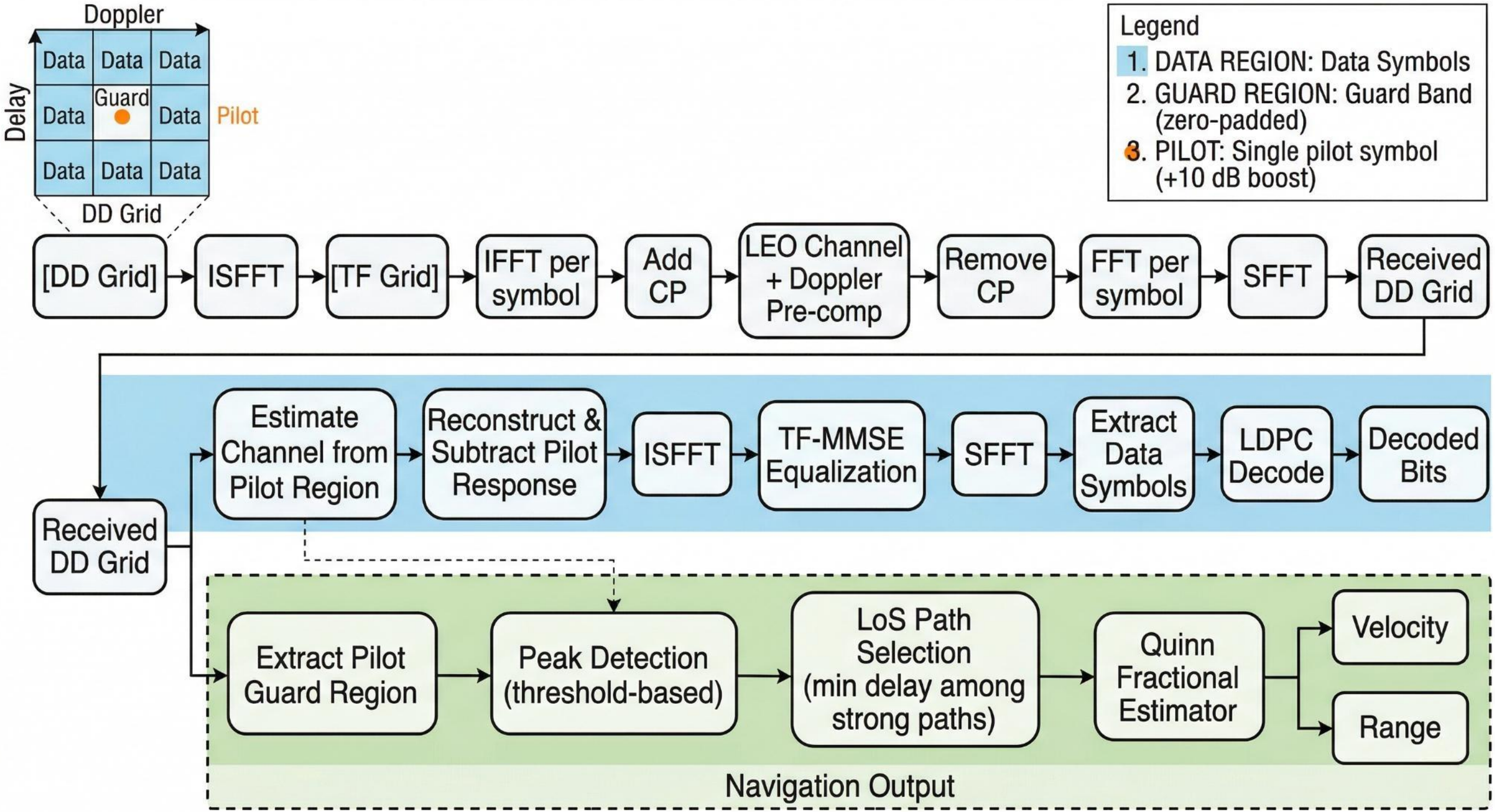
Doppler
Delay
Data
Guard
Pilot
DD Grid
Legend
1. DATA REGION: Data Symbols
2. GUARD REGION: Guard Band (zero-padded)
3. PILOT: Single pilot symbol (+10 dB boost)
[DD Grid]
ISFFT
[TF Grid]
IFFT per symbol
Add CP
LEO Channel + Doppler Pre-comp
Remove CP
FFT per symbol
SFFT
Received DD Grid
Received DD Grid
Estimate Channel from Pilot Region
Reconstruct & Subtract Pilot Response
ISFFT
TF-MMSE Equalization
SFFT
Extract Data Symbols
LDPC Decode
Decoded Bits
Extract Pilot Guard Region
Peak Detection (threshold-based)
LoS Path Selection (min delay among strong paths)
Quinn Fractional Estimator
Velocity
Range
Navigation Output